

Family 0.9.tf in two pages

family 1.0.tf with its raw imagery replaced by Google’s AlphaEarth embedding

blauewelt/earth — two-page summary of the design note, 17 September 2026. A glossary of every term follows the two pages.

What it is. Family 0.9.tf is family 1.0.tf (the land-and-coast stores) with **one substitution:** instead of the list of raw satellite and aerial images — and the small learned encoder that would turn them into tokens — it uses Google DeepMind’s published **AlphaEarth Satellite Embedding**: 64 numbers describing each 10 m of land, once a year, 2017–2025, computed from Sentinel-1, Sentinel-2 and Landsat and trained against radar, LiDAR, climate and land-cover sources; free under CC BY 4.0, 577 TB in all. **Everything else is inherited from 1.0.tf by reference** — the stations and gauges, the LiDAR, the daily fields, the prediction targets and the fixed maps — exactly as family 10.2 inherits family 10.1: the registry lists them with their 1.0.tf locations and checksums, and not one byte is rebuilt. **What it is for:** testing land prediction now, and measuring exactly whether the programme’s own image encoder is worth its compute — the two families differ only in the imagery. **What it is not:** not sub-annual (one vector per year, so a five-day event is averaged away), not extensible to new sensors (NISAR) or to years before 2017, and not a copy of Google’s 577 TB, which is read in place.

store	what it is	producer	space	time	record	stored size	track	phase
<i>built by this family: the AlphaEarth Satellite Embedding, v1 (CC BY 4.0)</i>								
aef1k	embedding averaged over each km ² , plus how uniform the cell is	Google DeepMind (AlphaEarth v1); pooled here	0.93 km (1/120°)	annual	2017–2025	≈ 0.2 TB (0.25 TB with high latitudes)	public	A
aef.dots	16 unaveraged 10 m vectors per 0.25° land cell per year	Google DeepMind (AlphaEarth v1)	10 m points	annual	2017–2025	7 GB	public	A
aef100r	embedding averaged over 100 m cells, chosen regions	Google DeepMind (AlphaEarth v1); pooled here	93 m (1/1200°)	annual	2017–2025	≈ 0.35 TB (frontiers); ≈ 5.7 TB (coastal band, C)	public	B / C
aef10	Google’s original 10 m files, read in place by byte range	Google Cloud Storage; Source Cooperative mirror	10 m	annual	2017–2025	0 (577 TB referenced)	public (external)	A
<i>inherited from 1.0.tf by reference: same files, same checksums, nothing rebuilt</i>								
ghcnd	daily temperature, rain and snow at weather stations	NOAA NCEI (GHCN-Daily)	point; 132,437 stations	daily	1763 →	≈ 40 GB (probe: 41 B/row)	<i>inherited from 1.0.tf</i> ; public	A
igra	weather-balloon soundings of the air column	NOAA NCEI (IGRA v2.2)	point; 2,932 sites	00 and 12 UTC	1905 →	7–9 GB (probe; note 10 GB)	<i>inherited from 1.0.tf</i> ; public	A
tide	hourly coastal sea level from tide gauges	GESLA-4 via UHSLC	point; 6,474 records	hourly or finer	1800s → 2026	40–50 GB (probe: 33 B/row)	<i>inherited from 1.0.tf</i> ; public	A
tide.private	tide gauges whose terms forbid republishing	GESLA-4 (CMEMS, CV, UZ; SANHO)	point	hourly or finer	as tide	10–20 GB [†] (probe: 33 B/row)	<i>inherited from 1.0.tf</i> ; private	A
ndbc	weather and waves at US buoys and coastal stations	NOAA NDBC	point; 1,345 stations	hourly; 10 min recently	1970 → 2025	≈ 30 GB (probe; note 15–25 GB)	<i>inherited from 1.0.tf</i> ; public	A
gliders	ocean profiles from autonomous underwater gliders	IOOS Glider DAC; OceanGliders	point profile	every ≈ 3 h	2003 →	0.8 GB (probe: 155 B/row)	<i>inherited from 1.0.tf</i> ; public	A
flux	land–air exchange of CO ₂ , water and heat	FLUXNET (AmeriFlux, ICOS, TERN, ChinaFLUX)	point; 700+ towers	half-hourly	1991 →	5 GB	<i>inherited from 1.0.tf</i> ; public	A
fire	satellite detections of active fires	NASA FIRMS (MODIS, VIIRS)	1 km / 375 m	per overpass	2000–11 →	20 GB	<i>inherited from 1.0.tf</i> ; public	A
river	daily river discharge (US and Caravan)	USGS NWIS; GRDC-Caravan	point; 24,976 + 5,356	daily	1800s →	14 GB	<i>inherited from 1.0.tf</i> ; public	B
river.grdc	daily river discharge worldwide	GRDC	point; > 10,000 gauges	daily	1800s →	≈ 5 GB [†]	<i>inherited from 1.0.tf</i> ; private	B
ismn	soil moisture and temperature in the ground	ISMN (TU Wien)	point; 3,300 stations	hourly	1952 →	≈ 15 GB [†]	<i>inherited from 1.0.tf</i> ; private	B
ghcnh	hourly weather-station reports	NOAA NCEI (GHCNh)	point; 38,870 stations	hourly / synoptic	1700s →	360 GB	<i>inherited from 1.0.tf</i> ; public	B
hfr	surface currents from shore-based radar	HFRNet (US); European HFR Node	6 / 2 / 1 km cells	hourly	US 2012 →; EU ≈2009 →	110–155 GB	<i>inherited from 1.0.tf</i> ; public	B
alticap	satellite sea level close to the coast	ALTICAP (AVISO/ODATIS) Jason-3	≈ 350 m long track	≈ 10-day repeat	2016-02 → 2021-07	3 GB	<i>inherited from 1.0.tf</i> ; public	B
ecad	European station weather missing from GHCN-Daily	ECA&D (KNMI)	point	daily	1800s →	≤ 20 GB [†]	<i>inherited from 1.0.tf</i> ; private	C
gedi	laser canopy height, cover and biomass	NASA GEDI (LP DAAC, ORNL)	25 m footprints	per orbit pass	2019-04 → 2023-03; 2024-04 → 2025-07	0.2–0.5 TB	<i>inherited from 1.0.tf</i> ; public	B

store	what it is	producer	space	time	record	stored size	track	phase
icesat2	laser terrain, canopy and sea-ice height	NASA ICESat-2 (NSIDC)	100 m segments (≈ 11 m spots)	per orbit pass	2018-10 $\rightarrow$	≈ 1.1 TB	<i>inherited from 1.0.tf</i> ; public	C
1st05	land-surface temperature, day and night (Terra in phase A)	NASA MODIS Terra (+ Aqua, VIIRS)	5.6 km (0.05°)	daily	2000-02 $\rightarrow$	≈ 350 GB per satellite	<i>inherited from 1.0.tf</i> ; public	A / B
snow05	snow-cover fraction	NASA MODIS (NSIDC)	0.05°	daily	2000-02 $\rightarrow$	≈ 60 GB	<i>inherited from 1.0.tf</i> ; public	A
refl05	surface reflectance: the colour of the ground	NASA MODIS (+ VIIRS)	0.05°	daily	2000-02 $\rightarrow$	≈ 700 GB (+330 GB VIIRS)	<i>inherited from 1.0.tf</i> ; public	B
ims1k	analyst-drawn snow and ice chart	NOAA IMS (NSIDC)	1 km (4 km and 24 km earlier)	daily	1 km 2014-12 $\rightarrow$; 4 km 2004, 24 km 1997	≈ 50 GB	<i>inherited from 1.0.tf</i> ; public	B
chirps05 (E4)	rainfall from satellite infrared and gauges	UCSB CHC (CHIRPS v3.0)	0.05°	pentad (6 a month)	1981 $\rightarrow$	≈ 40 GB	<i>inherited from 1.0.tf</i> ; public	B
alerts	dated forest-clearing alerts	UMD GLAD, WUR RADD, NASA OPERA	10–30 m	per detection	2015 $\rightarrow$	≈ 30 GB	<i>inherited from 1.0.tf</i> ; public	A
lossyear	year each 30 m pixel lost its forest	UMD Hansen GFC; EU JRC TMF	30 m	annual	1990 / 2001 $\rightarrow$ 2025	≈ 30 GB	<i>inherited from 1.0.tf</i> ; public	A
burned500	date each pixel burned	NASA MODIS MCD64A1	500 m	monthly	2000-11 $\rightarrow$	≈ 10 GB	<i>inherited from 1.0.tf</i> ; public	A
cat_gfm	list of flood and surface-water maps	Copernicus GFM; NASA OPERA DSWx	20–30 m	per pass	2015 / 2016 $\rightarrow$	< 1 GB	<i>inherited from 1.0.tf</i> ; public	A
biomass100	yearly above-ground biomass	ESA CCI Biomass; GEDI L4B	100 m (GEDI 1 km)	annual	2005–2012, 2015–2024	≈ 0.4 TB	<i>inherited from 1.0.tf</i> ; public	C
landcover10	yearly land-cover and crop-type maps	Esri/IO, ESA, MapBiomas, USDA, JRC	10–30 m	annual	1985 / 2017 $\rightarrow$	≈ 0.2 TB [†]	<i>inherited from 1.0.tf</i> ; public	C
static_fine	fixed maps: elevation and depth, land cover, distance to coast	GEBCO, ESA WorldCover, GSHHG	15'' (≈ 460 m); 0.05°	fixed	—	≈ 8 GB	<i>inherited from 1.0.tf</i> ; public	A

Track and phase of an inherited store are 1.0.tf's. The ten image catalogues of 1.0.tf (`cat_landsat`, `cat_hls`, `cat_s2`, `cat_s1`, `cat_nisar`, `cat_palsar`, `cat_olci`, `cat_viirs`, `cat_ecostress`, `cat_submetre`), the tile tokens and `oc300_coast` are **not** inherited: the four AlphaEarth stores replace them. Inherited sizes and probes are 1.0.tf's (**Probe**: a one-month test fetch, E-082); [†] estimate; B = bytes.

How it is stored. The registry `tensors/family09.tf/family09tf.json` carries an `inherits` block naming family 1.0.tf's registry; each inherited store keeps its 1.0.tf path and checksums (the four private ones stay in the private repository), and only the four AlphaEarth groups live under `family09.tf`. The pooled stores (`aef1k`, `aef100r`) average the unit-length 10 m vectors in each cell, renormalise, and keep the average's length as a 65th *coherence* channel (1 = the cell is one thing, near 0 = many), all in one byte per number, as compressed tiles; `aef_dots` keeps raw vectors as point rows; `aef10` points at Google's files. A year- Y vector is filed in the first five-day bin after 31 December Y , so a prediction inside year Y never sees that year's summary.

The ladder

- A** ≈ 0.2 TB: `aef1k`, `aef_dots`, the `aef10` reference and the registry with its `inherits` block; built in one pass next to the mirror, one lane per year, \$50–150. Inherited stores appear as 1.0.tf publishes them (its phases: A ≈ 0.7 TB, B 2.3–2.7 TB, C ≈ 3 TB).
- B** `aef100r` over three deforestation frontiers (Amazon, Congo, insular Southeast Asia), ≈ 0.35 TB.
- C** `aef100r` over the ± 100 km coastal band (≈ 5.7 TB), once E-078c's coastal cells exist to compare against.

Glossary

Every term, acronym, producer, instrument and code name used in this summary and in the full design note, in alphabetical order.

aef10 — Family 0.9.tf registry group pointing at Google’s original 10 m files, read in place.

aef100r — Family 0.9.tf store of the embedding averaged over ≈ 100 m cells in chosen regions.

aef1k — Family 0.9.tf store of the embedding averaged over each ≈ 1 km cell, with a coherence channel.

AEF band names A00–A63 — The names of the 64 embedding values in Earth Engine.

aef_custom — A possible private store of sub-annual AlphaEarth embeddings, only if Google’s academic programme grants a sample.

aef_dots — Family 0.9.tf store of 16 unaveraged 10 m embedding vectors per 0.25° land cell per year, at the same positions every year.

aef_index.parquet — Google’s 70 MB table of every embedding tile.

alerts — Family 1.0.tf target store of dated forest-disturbance alerts (GLAD, RADD, OPERA) as point rows.

ALOS, PALSAR, PALSAR-2 — Japan’s Advanced Land Observing Satellites and their L-band radars.

AlphaEarth Foundations (AEF) — Google DeepMind’s model that turns sequences of satellite images into one 64-number description of each 10 m of land for each year; its published output is the *Satellite Embedding*.

ALTICAP — A coastal altimetry product that reprocesses Jason-3 to within a few hundred metres of the shore.

alticap — Family 1.0.tf store of ALTICAP coastal sea level along Jason-3 tracks, 2016–2021.

altimeter, along-track — A radar satellite that measures sea-surface height straight below it; along-track data are its measurements along that ground line.

AmeriFlux, ICOS, TERN, ChinaFLUX, FLUXNET — The flux-tower networks of the Americas, Europe, Australia and China, and FLUXNET, their global collection (the FLUXNET Shuttle release).

anchor — The place and time the model is asked to predict; everything it reads is placed relative to it.

annual target — A yearly map the model is asked to predict (a loss year, a biomass map); it is filed at the end of its year so it is never read as an input from inside that year.

attribution, yes, no (licence.redistribution) — Registry values saying whether a licence lets the data be republished freely, with credit, or not at all.

AVISO, ODATIS — French services that distribute satellite altimetry and ocean data.

AWS, AWS Open Data — Amazon Web Services, and its programme that hosts public datasets at no cost to readers.

bin (pentad) — The five-day time step every family uses, counted from 1 January 1982; a bin before 1982 has a negative number.

biomass100 — Family 1.0.tf target store of yearly above-ground biomass maps at 100 m.

box (rented, verified) — A machine hired by the hour from a marketplace; it holds no credentials, and a verified one runs on a host the marketplace has vetted.

B/row (bytes per row) — Stored size of one row of a point store; a row costs $27 + 2$ bytes per channel ($31 + 2$ per channel with the 64-bit time).

burned500 — Family 1.0.tf target store of the date each 500 m pixel burned.

Caravan — An open (CC BY 4.0) collection of catchment discharge and weather data, including a GRDC subset.

cat_ecostress — Catalogue of ECOSTRESS 70 m thermal scenes.

cat_gfm — Target catalogue of flood and surface-water maps (Copernicus GFM, OPERA DSWx); a read-out, not imagery.

cat_hls — Catalogue of NASA’s harmonised Landsat and Sentinel-2 scenes.

cat_landsat — Catalogue of every Landsat scene since 1982, all sensors.

cat_nisar — Catalogue of NISAR L-band radar scenes.

cat_olci — Catalogue of Sentinel-3 OLCI 300 m ocean-colour scenes over the coastal band.

cat_palsar — Catalogue of JAXA’s PALSAR yearly radar mosaics and wide-swath scenes (exception E5).

cat_s1 — Catalogue of Sentinel-1 radar scenes.

cat_s2 — Catalogue of Sentinel-2 optical scenes.

cat_submetre — Catalogue of sub-metre and metre-scale aerial and basemap imagery where it is openly published (exception E6).

cat_viirs — Catalogue of VIIRS 375 m imager scenes.

C band, L band — Radar wavelengths of about 5.5 cm and 24 cm; L band sees deeper into vegetation.

CC0, CC BY, CC BY-NC — Creative Commons licences: public domain; free with attribution; free with attribution for non-commercial use only.

C (channels) — The number of values stored per row or per pixel.

checksum (sha256) — A 64-character fingerprint of a file; if one byte changes, the fingerprint changes.

chfrank/earth-tensors, chfrank/earth-tensors-private — The project’s public and private Hugging Face dataset reposi-

tories.

CHIRPS — The Climate Hazards Center InfraRed Precipitation with Station data rainfall record (from 1981).

chirps05 — Family 1.0.tf store of CHIRPS five-day rainfall at 0.05° (exception E4), with its **frame_table** of frame dates.

CMEMS, CV, UZ — Three GESLA contributor codes whose tide-gauge data are for research use only.

codec (local codec) — A small learned encoder that turns an image tile into a few compact vectors (tokens) the model can read.

COG — Cloud-optimised GeoTIFF: an image file laid out so any part of it can be read by byte range over HTTP.

coherence — In **aef1k**, the length of the average of a cell's unit vectors: 1 if its pixels agree, near 0 if they differ.

companding (int8) — Storing a value in one signed byte on a square-root scale, which keeps more precision near zero; AlphaEarth's files use it.

cone — The set of past measurements the model reads around the anchor: wider for older times, like a cone in space and time.

Copernicus, Copernicus Marine — The EU's Earth-observation programme and its marine service (free account required).

Copernicus DEM, GLO-30 — The EU's global 30 m elevation model.

Custom Satellite Embeddings — Google Maps Platform's sub-annual embeddings, in private preview with no public licence.

DAAC (LP, ORNL, PO, OB) — NASA's discipline data centres: LP DAAC for land, ORNL DAAC for ecology, PO.DAAC for oceans, OB.DAAC for ocean biology.

DAC, GDAC — Data assembly centre, and the global one that gathers a network's data (Argo, gliders, OceanSITES).

DeepMind — Google's AI research laboratory, co-producer of AlphaEarth.

distribution — Registry field naming a store's track, **public** or **private**.

done.json — A marker a build writes last, so an unfinished build can never look finished.

DSWx-HLS, DSWx-S1 — OPERA's 30m surface-water extent maps from HLS optical and from Sentinel-1 radar.

E-078, E-078c — The design study that gave family 10 one schema for every granularity, and its first experiment training a local image codec on about 300 coastal cells.

E-082 — The build plan for families 1.gf, 1.0.tf and 0.9.tf; its first wave wrote and probed the phase-A point-store adapters.

E1–E6 (exceptions) — Sources admitted although they break the 10 km / 5-day rule: E1 scatterometer wind at 12.5 km, E2 IMERG rain at 11 km, E3 GOSAT at 10.5 km, E4 CHIRPS rain, E5 PALSAR yearly mosaics, E6 sub-metre photos.

Earth Engine — Google's hosted geospatial analysis platform (account required).

ecad — Private family 1.0.tf store of ECA&D European station series missing from GHCN-Daily.

ECA&D; TX, TN, TG, RR, SS, SD, HU, PP — The European Climate Assessment & Dataset of station series, and its element codes: maximum, minimum and mean temperature, rain, sunshine, snow depth, humidity and pressure.

ECMWF — The European Centre for Medium-Range Weather Forecasts, producer of ERA5.

embedding — A learned list of numbers summarising what was observed at a place, arranged so that similar places get similar numbers.

ERA5, ERA5-Land — ECMWF's global reanalysis and its land-surface companion.

ESA — The European Space Agency.

ESA CCI — ESA's Climate Change Initiative, which builds long consistent climate records from satellites.

Esri, Impact Observatory — The companies that produce a yearly 10 m global land-cover map.

falsifier — The test, fixed before building, that can show a store is not worth keeping.

family — A numbered set of datasets built to one storage design and read by the programme's forecasting model.

family 0.9.tf — Family 1.0.tf with its raw-imagery catalogue replaced by Google's AlphaEarth embedding, everything else inherited.

family 10, 10.1, 10.2 — The granularity-aware dataset set in which every measurement keeps its own place, time and footprint; 10.1 built four point stores (drifters, tropical moorings, ship CO₂, satellite sea level) and 10.2 inherited them and added fishing effort.

family10.json, family1gf.json, family1tf.json, family09tf.json — The registry files of families 10, 1.gf, 1.0.tf and 0.9.tf.

family 1.0.tf — The proposed land-and-coast family at the same scale, built from raw observations and a catalogue of raw imagery ("tf": terrestrial, fine).

family 1.gf — The proposed global family of ocean and atmosphere measurements finer than 10 km and 5 days ("gf": global, fine).

family 7, 7.1, 7.2 — The published global input tensor: every 0.25° cell from pole to pole in five-day steps, 1982–2024; 7.1 added ocean colour and 7.2 is the current version.

fire — Family 1.0.tf store of FIRMS satellite fire detections.

FIRMS; FRP — NASA's Fire Information for Resource Management System, and fire radiative power, the energy a fire gives off.

flux — Family 1.0.tf store of half-hourly flux-tower measurements.

footprint; $\log_2 f_p$, $\log_2 f_t$ (log2_fp, log2_dt, fp) — The area and time span a value averages over, stored as base-2 logarithms relative to 27.83 km and 5 days; -4 is the label for a point measurement with no area.

frame; **frames_per_bin**, **frame_seconds**, F — One time slice of a gridded field; a five-day bin holds F of them (5 for daily, 40 for three-hourly), each lasting **frame_seconds**.

GBIF — The Global Biodiversity Information Facility, an open database of species occurrences.

GCS (**gs://**) — Google Cloud Storage, Google’s object store; **gs://** addresses a file in it.

GDAL, **VRT** — GDAL is the standard open library for geospatial rasters; a VRT is its small text file that describes how to read another raster.

GEBCO — The General Bathymetric Chart of the Oceans, a global elevation and sea-floor-depth grid.

gedi — Family 1.0.tf store of GEDI laser footprints: canopy height profile, cover and biomass.

GEDI; **L2A**, **L2B**, **L4A**, **L4B** — The Global Ecosystem Dynamics Investigation, a laser on the Space Station measuring canopy height; its products for height profiles, cover, footprint biomass and gridded biomass.

GESLA, **GESLA-4** — The Global Extreme Sea Level Analysis compilation of hourly tide-gauge records, version 4.

GFM — Copernicus’ Global Flood Monitoring service, flood extent from every Sentinel-1 pass.

ghcnd — Family 1.0.tf store of GHCN-Daily station records.

GHCN-Daily, **GHCNh**, **ISD** — NOAA’s Global Historical Climatology Network of daily and hourly station records; **GHCNh** replaces the frozen Integrated Surface Database.

ghcnh — Family 1.0.tf store of GHCN-hourly station reports.

GLAD, **GLAD-L**, **GLAD-S2** — The University of Maryland’s forest-loss alerts from Landsat and from Sentinel-2.

gliders — Family 1.0.tf store of glider profiles at 16 depths.

GRACE — Satellites that measure changes in Earth’s gravity, and so in stored water and ice.

GRDC — The Global Runoff Data Centre’s archive of river discharge from more than 10,000 gauges.

GSHHG — The Global Self-consistent Hierarchical High-resolution Geography shoreline set (public domain).

Hansen GFC — The University of Maryland’s Global Forest Change map of tree cover in 2000 and the year of loss since.

hfr — Family 1.0.tf store of hourly surface currents from shore-based radar.

HFR, **HFRNet** — High-frequency radar that measures surface currents from the shore, and the US network of such radars (with a European node).

HTTP, **HTTPS**, **HTTP range** — The web’s transfer protocol (HTTPS is the encrypted form); a range request asks for one byte span of a file.

Hub (Hugging Face) — Hugging Face’s dataset hosting, where the project publishes its stores; it limits files per repository and per folder.

ICESat-2 — NASA’s polar laser-altimetry satellite (2018 on).

icesat2 — Family 1.0.tf store of ICESat-2 terrain, canopy and sea-ice heights per 100 m segment.

IGRA — The Integrated Global Radiosonde Archive of weather-balloon soundings.

igra — Family 1.0.tf store of IGRA radiosonde soundings at 16 standard levels.

IMS — NOAA’s Interactive Multisensor Snow and Ice Mapping System, a daily analyst-drawn chart.

ims1k — Family 1.0.tf store of the daily 1 km IMS snow and ice chart.

inherits — Registry block naming another family’s registry and the groups taken from it unchanged, each with its original path and checksums.

IOOS — The US Integrated Ocean Observing System, which runs the national glider data centre.

ISMN — The International Soil Moisture Network of in-ground sensors.

ismn — Private family 1.0.tf store of ISMN soil moisture and soil temperature.

ISRO — The Indian Space Research Organisation, NASA’s partner on NISAR.

Jason-3 — A radar altimeter satellite (2016 on) measuring sea level.

JAXA — Japan’s space agency.

JRC — The European Commission’s Joint Research Centre.

JRC TMF — The JRC’s Tropical Moist Forest map of yearly forest change since 1990.

KB, **MB**, **GB**, **TB**, **PB** — Kilo-, mega-, giga-, tera- and petabytes: each a thousand times the last.

KNMI — The Royal Netherlands Meteorological Institute, which runs ECA&D.

landcover10 — Family 1.0.tf target store of yearly land-cover and crop-type maps.

Landsat 4, **5**, **7**, **8**, **9**; **TM**, **ETM+**, **OLI**, **TIRS** — The US optical Earth-observation satellites since 1982, and their cameras: Thematic Mapper, Enhanced Thematic Mapper Plus, Operational Land Imager and Thermal Infrared Sensor; Collection 2 is the current reprocessing.

lat, **lon**, **values**, **qc**, **platform**, **time.s** — Arrays of a point store: position, the measured values, a quality code, a hashed instrument identity, and the time in seconds since 1982.

layout, **prefix**, **path**, **cadence**, **footprint**, **tier** — Registry fields per group: storage layout, folder, the group’s own location, time step, footprint and tier.

LiDAR — Laser ranging: a pulse’s return time gives the height of the ground and of each layer of a canopy.

lossyear — Family 1.0.tf target store of the year each 30 m pixel lost its forest.

1st05 — Family 1.0.tf store of daily 0.05° land-surface temperature, day and night.

Major-TOM — An open Earth-observation dataset project whose Hugging Face sample of the AlphaEarth embedding is used as a test fixture.

MapBiomass — Brazil’s annual 30 m land-cover mapping project.

MOD09CMG, MOD10C1, MOD11C1, MYD11C1, MCD64A1 — MODIS product names: daily 0.05° reflectance, snow cover and land-surface temperature (MOD for Terra, MYD for Aqua), and monthly burned area (MCD, both).

MODIS; Terra, Aqua — The Moderate Resolution Imaging Spectroradiometer on NASA’s Terra (1999) and Aqua (2002) satellites, both near the end of their lives.

NaN — “Not a number”, the stored value for “not measured”.

NASA — The US space agency.

NCEI, OCADS — NOAA’s National Centers for Environmental Information, the US national environmental archive, and its ocean-carbon archive.

NDBC — NOAA’s National Data Buoy Center.

ndbc — Family 1.0.tf store of NDBC buoy and coastal-station weather and waves.

NISAR; GCOV — The NASA–ISRO L-band radar satellite (launched 2025), and its geocoded covariance product.

NLCD — The US National Land Cover Database.

NOAA, NOAA STAR — The US National Oceanic and Atmospheric Administration, and its satellite research centre.

nodata — The marker value a file uses for “no measurement here”.

n_R — The number of readings a token carries; $n_R = 0$ says “nothing was measured here”.

NSIDC — The US National Snow and Ice Data Center.

NWIS — The USGS National Water Information System, the US river-gauge archive.

oc300_coast — A possible phase-C gridded store of 300 m OLCI ocean colour over the coastal band.

OceanGliders — The international glider programme and its data centre.

OPERA — NASA’s Observational Products for End-Users from Remote Sensing Analysis: disturbance and surface-water maps from HLS and Sentinel-1.

orthophoto — An aerial photograph corrected so that it lines up with a map.

phase A, B, C — The three build waves of each family: A cheap and first, B the terabyte stores, C decisions taken after A and B have verdicts.

probe — A one-month (or one-zone) test run of a store’s real fetch code, measuring rows, bytes and speed before any full build.

public track, private track — Where a store lives: the public Hugging Face repository that anyone can read, or a private one used only by this project because the source’s licence forbids republishing.

RADD — Radar for Detecting Deforestation, Wageningen’s Sentinel-1 forest-loss alerts.

radiosonde — An instrument package on a weather balloon that measures temperature, humidity and wind as it rises.

read-out — A quantity the model is asked to predict or report, such as the Niño-3.4 index or next year’s forest loss.

reanalysis — A weather or ocean model run over past decades and continually corrected towards observations, giving complete fields that are partly model.

ref105 — Family 1.0.tf store of daily 0.05° land-surface reflectance.

registry — The JSON file listing every store of a family with its tier, location, channels, checksums and licence; the one file a reader opens first.

river — Family 1.0.tf store of public daily river discharge (USGS and Caravan).

river_grdc — Private family 1.0.tf store of GRDC daily river discharge.

S3 (s3://) — Amazon’s object-storage service and its address form; many archives serve files through it.

SANHO — The South African Navy Hydrographic Office.

schema 2, schema 3 — The column layout of a point store: schema 3 widens the time column to 64 bits so records before 1914 fit.

seed — One independent training run; three seeds means the same test run three times with different random starts.

Sentinel-1, -2, -3, -5P, -6A (S1A, S1C, S1D, S2A, S2B, S2C) — The EU Copernicus satellites: C-band radar, 10 m optical, ocean colour and temperature, trace gases, and sea-level altimetry; the letters are individual satellites of a series.

shard, tile; shard_index.npy, tile_grid.json, n_tiles — In the G-fine layout, one compressed file per group and bin, cut into 256×256 -pixel tiles; the index gives each tile’s byte position and the grid file each tile’s corners.

siblings — A list in family 10.2’s registry naming the registries of families 1.gf, 1.0.tf and 0.9.tf.

slatrack — Family 10.1’s store of satellite sea level along the ground track, all missions (67 GB).

snow05 — Family 1.0.tf store of daily 0.05° snow cover.

Source Cooperative (tge-labs/aef) — A public data host where the Taylor Geospatial Engine keeps a complete mirror of the AlphaEarth embedding.

spot instance — A discounted cloud machine that the provider may reclaim at short notice.

SST — Sea-surface temperature.

static.fine — Family 1.0.tf store of fixed fields: elevation and bathymetry, land-cover fractions, distance to coast.

store — One dataset of a family, published as a small set of arrays plus a description file.

support period, valid period — The dates of the images an AlphaEarth vector was computed from, and the period it describes (one calendar year in the published product).

tide — Family 1.0.tf store of public GESLA tide-gauge sea level.

tide_private — Private sibling of **tide** holding the gauges whose terms forbid republishing.

tier G-fine (sharded) — The layout family 1.gf adds for grids finer than 0.25°: each five-day bin is one compressed file cut into tiles, so a read fetches only the tiles it needs.

tier G, P, T — Family 10's three storage forms: G, gridded fields read one time step at a time; P, points, profiles and tracks read by nearest-neighbour search; T, image tiles held as a catalogue plus codec tokens.

tile tokens — The codec's vectors for each image tile, a later store sized by E-078c.

token — The unit the model reads: a value, its offset in space and time from the anchor, and its footprint.

TROPOMI — The trace-gas imager on Sentinel-5P, measuring NO₂, CO and CH₄ (nitrogen dioxide, carbon monoxide, methane).

TU Wien — The Technical University of Vienna, which hosts ISMN.

UCSB CHC — The Climate Hazards Center at the University of California, Santa Barbara, producer of CHIRPS.

UHSLC — The University of Hawaii Sea Level Center.

UK, US, EU — United Kingdom, United States, European Union.

UMD — The University of Maryland, producer of GLAD and the Hansen forest-change map.

unit length, S^{63} — AlphaEarth's vectors all have length one, so they lie on a sphere in 64 dimensions and their dot product measures similarity.

URL — A web address.

USDA — The US Department of Agriculture.

USGS — The US Geological Survey.

UTC — Coordinated Universal Time, the time zone of weather observations.

UTM — Universal Transverse Mercator: sixty map zones that each keep a flat metre grid; AlphaEarth's tiles use them.

VIIRS; S-NPP, NOAA-20, NOAA-21 — The Visible Infrared Imaging Radiometer Suite, MODIS's successor, on the Suomi-NPP (ending 2026-11), NOAA-20 and NOAA-21 satellites; I-band is its 375 m channel set.

WorldCover — ESA's 10 m global land-cover map (2020 and 2021).

WUR — Wageningen University & Research, producer of RADD.